7	(a)		$\theta = \frac{72.576}{r^2}$ or $72.576\ r^{-2}$ isw	**B1** **[1]**	**1.1**	eg $\frac{9072}{125r^2}$ or $\frac{9072r^{-2}}{125}$	
7	(b)		$r\theta + 2r$ or $r(\theta + 2) = 24.48$ seen **and** $\theta = \frac{24.48 - 2r}{r}$ or equivalent constructive step to give $\frac{24.48}{r} - 2$ **AG**	**B1** **[1]**	**1.1**	or $2\pi r\frac{\theta}{2\pi} + 2r = 24.48$	

Question			Answer	Marks	AOs		Guidance
7	(c)		$their\ \frac{2\times36.288}{r^2}=\frac{24.48}{r}-2$ oe	M1	3.1a	NB 72.576	
			$r^2-12.24r+36.288$ [= 0]	M1	2.1	quadratic obtained in form f(r) [= 0]	
			[r =] 5.04 or 7.2 oe	A1	1.1		allow **B3** for 5.04 **and** 7.2 unsupported **or** allow **SC3** for obtaining and solving an equation for θ and then finding both values of r
				[3]			